

Multicast Attack Rule-Based Detection

IGMP & PIM Security Monitoring Framework

This document explains the logic, detection methodology, protocol handling, and anomaly detection techniques used in the multicast security detection engine implemented for:

- Internet Group Management Protocol (IGMP)
- Protocol Independent Multicast (PIM)

The implementation focuses on identifying multicast control-plane abuse, flooding attacks, spoofing attempts, and multicast reconnaissance behavior from PCAP traffic captures.

The system is designed for lightweight real-time behavioral analysis using protocol-aware heuristics instead of signature-only detection.

1. IGMP Attack Detection Engine

Source implementation:

1.1 Objectives

The IGMP detection engine is designed to detect:

Attack Type	Purpose
IGMP General Query Flood	Router CPU/resource exhaustion
IGMP Group-Specific Query Flood	Excessive multicast query abuse
Multicast Group Scanning	Enumeration of multicast groups
Unknown Querier Activity	Spoofed multicast router behavior
TTL Violations	Off-link spoofed multicast traffic

1.2 Protocol Background

IGMP is used between hosts and multicast routers for multicast group membership management.

Important IGMP Message Types

Type	Hex	Purpose
Membership Query	0x11	Router asks hosts about memberships
IGMPv1 Report	0x12	Host joins multicast group
IGMPv2 Leave	0x17	Host leaves multicast group

Type	Hex	Purpose
IGMPv3 Report	0x22	Advanced multicast membership report

1.3 IGMP Version Handling

The detector supports IGMPv1, IGMPv2, and IGMPv3.

Version Classification Logic

```
def classify_igmp_version(self, igmp_type, version_hint=None):
```

Detection Mapping

IGMP Type Detected Version

0x12 IGMPv1

0x17 IGMPv2

0x22 IGMPv3

Otherwise Default IGMPv2

Why This Matters

Different IGMP versions behave differently:

- IGMPv1 uses only membership reports
- IGMPv2 introduces leave messages
- IGMPv3 supports source filtering and advanced multicast control

The detector adapts automatically based on observed packet types.

1.4 General Query Flood Detection

Attack Description

An attacker floods the network with excessive IGMP General Queries.

Normal multicast routers send queries periodically (typically every 125 seconds).

Flooding queries can:

- overload hosts
 - increase multicast processing
 - destabilize multicast routing
 - trigger unnecessary membership reports
-

Detection Logic

Step 1 — Identify General Query

`igmp_type == 0x11 and group == "0.0.0.0"`

A General Query always targets 0.0.0.0.

Step 2 — Measure Inter-Arrival Time (IAT)

`iat = ts - self.last_general_query_time`

If packets arrive too quickly:

`iat < GENERAL_QUERY_IAT_THRESHOLD`

the behavior becomes suspicious.

Step 3 — Burst Tracking

The detector stores timestamps inside:

`self.general_query_history`

using a sliding time window.

Step 4 — Flood Decision

If the number of queries exceeds the threshold:

`len(self.general_query_history)`

`>= GENERAL_QUERY_BURST_THRESHOLD`

an alert reason is generated:

"General Query Flood"

1.5 Group-Specific Query Flood Detection

Attack Description

Instead of flooding general queries, attackers repeatedly query specific multicast groups.

Purpose:

- target multicast applications
- disrupt IPTV/video streaming
- overload multicast routers
- force repeated group state processing

Detection Logic

Step 1 — Identify Group-Specific Query

`igmp_type == 0x11 and group != "0.0.0.0"`

Step 2 — Track Queries Per Group

Each multicast group has its own deque:

`self.group_query_history[group]`

Step 3 — Sliding Window Analysis

The detector keeps only recent timestamps:

`GROUP_QUERY_WINDOW`

Step 4 — Flood Threshold

If queries exceed:

`GROUP_QUERY_BURST_THRESHOLD`

the group becomes suspicious.

1.6 False Positive Reduction Using Membership Correlation

One important feature of the detector is behavioral correlation.

The detector verifies whether a legitimate membership event recently occurred.

Member Events Recorded

Packet Type	Meaning
IGMPv2 Leave	Host changed membership
IGMPv3 Report	Host membership update

Logic

If a multicast membership event happened recently:

`has_recent_member_event()`

the query flood may actually be legitimate.

This reduces false positives during:

- real multicast joins
 - channel switching
 - IPTV group changes
-

1.7 Multicast Group Scan Detection

Attack Description

An attacker probes many multicast groups rapidly.

Purpose:

- multicast reconnaissance
 - IPTV discovery
 - multicast application enumeration
 - topology mapping
-

Detection Logic

Step 1 — Track Unique Groups Per Source

```
self.group_scan_groups[src_ip][group] = ts
```

Step 2 — Maintain Time Window

Old entries are removed using:

```
GROUP_SCAN_WINDOW
```

Step 3 — Count Unique Groups

If the number of distinct multicast groups exceeds:

```
MAX_UNIQUE_GROUPS
```

the source is flagged as:

```
"Multicast Group Scan"
```

1.8 Unknown Querier Detection

Problem

Legitimate IGMP queries should normally come from stable multicast routers.

Attackers may spoof router behavior.

Detection Logic

Router Maturity Model

A router becomes trusted only after being observed continuously for:

QUERIER_MATURITY_SEC

Unknown Router Detection

If a source sends queries before becoming trusted:

src_ip not in self.known_queriers

the detector raises:

"Unknown Querier"

1.9 TTL Validation

Why TTL Matters

IGMP packets should never cross routers.

RFC-compliant IGMP packets use:

TTL = 1

Detection Logic

if ttl != 1:

Alert generated:

"Invalid TTL"

Security Benefit

Detects:

- off-link spoofing
 - malformed traffic
 - replayed multicast packets
 - tunneling abuse
-

1.10 Multi-Factor Suspicion Scoring

The detector avoids triggering alerts from a single anomaly.

Instead:

if suspicion ≥ 2 :

An alert is generated only when multiple suspicious behaviors occur together.

Benefits

Reduces:

- false positives
- noisy alerts
- transient multicast anomalies

Improves:

- behavioral accuracy
 - attack confidence
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1.11 PCAP Processing Pipeline

The detector processes packets as follows:

PCAP → IP Parsing → IGMP Validation →

Packet Classification → Stateful Analysis →

Rule Evaluation → Alert Generation

IGMP Packet Validation

The engine validates:

- IP layer exists
 - protocol number = 2
 - minimum IGMP payload length
-

2. PIM Hello Manipulation Detection Engine

Source implementation:

2.1 Objectives

The PIM detector identifies:

Attack	Purpose
PIM Hello Flood	Router CPU exhaustion
Fake Neighbor Injection	Rogue multicast router creation
Neighbor Explosion	Multicast topology instability
DR Priority Hijacking	Rogue Designated Router election
Holdtime Manipulation	Neighbor instability
Distributed Spoofed Neighbor Attack	Large-scale spoofing
Invalid TTL	Off-link spoofing
Invalid PIM Version	Malformed traffic
PIM on Access Port	Protocol misuse

2.2 PIM Protocol Background

PIM is used between multicast routers for multicast routing.

PIM Hello Messages

PIM Hello packets:

- establish neighbors
 - maintain adjacencies
 - elect Designated Routers (DR)
 - exchange capabilities
-

2.3 PIM Hello Flood Detection

Attack Description

An attacker sends excessive PIM Hello packets.

Effects:

- CPU exhaustion
 - neighbor churn
 - routing instability
-

Detection Logic

Count Hellos Per Router

```
self.hello_history[src_ip]
```

Sliding Window Tracking

Recent Hello timestamps are maintained inside:

WINDOW_SEC

Flood Detection

If count exceeds:

MAX_HELLO_COUNT

alert reason:

"Excessive Hellos"

2.4 Very Short Hello Interval Detection

Attack Description

Attackers may rapidly transmit Hellos at abnormal intervals.

Detection Logic

Inter-arrival time:

$iat = q[-1] - q[-2]$

If:

$iat < MIN_HELLO_INTERVAL$

detector raises:

"Very short hello interval"

2.5 Fake Neighbor Injection

Attack Description

Attackers create fake multicast neighbors using spoofed IPs.

Detection Logic

New neighbors are tracked using:

`self.first_seen`

and:

`self.new_neighbor_history`

Distributed Attack Detection

If too many new neighbors appear within:

`NEW_NEIGHBOR_WINDOW`

detector raises:

"Rapid fake neighbor creation"

2.6 Neighbor Explosion Detection

Attack Description

Large numbers of multicast neighbors suddenly appear.

Detection Logic

Current active neighbors:

`self.last_seen`

If neighbor count exceeds:

`MAX_NEIGHBORS`

alert:

"Neighbor explosion"

2.7 DR Priority Hijacking

Attack Description

Attackers manipulate Designated Router elections.

Higher DR priority can force attacker-controlled routers to become multicast DR.

Detection Logic

DR priority values are tracked:

```
self.dr_priority_history[src_ip]
```

If:

```
dr_priority > MAX_DR_PRIORITY
```

alert generated:

```
"High DR priority"
```

2.8 Holdtime Manipulation

Attack Description

Very low holdtimes cause frequent neighbor expiration.

Effects:

- instability
 - adjacency resets
 - multicast churn
-

Detection Logic

if holdtime < MIN_HOLDTIME:

alert:

```
"Low holdtime"
```

2.9 Invalid TTL Detection

Like IGMP, PIM Hello packets should remain local.

Expected:

```
TTL = 1
```

Any deviation indicates:

- spoofing
 - tunneling
 - malformed traffic
-

2.10 Invalid PIM Version Detection

The implementation validates:

if pim_version != 2:

Unexpected versions are suspicious because modern deployments use PIMv2.

2.11 Access Port PIM Detection

Problem

PIM should normally appear only on router-facing interfaces.

Detection Logic

interface.startswith("access")

If true:

"PIM seen on access port"

2.12 Unknown Router Burst Detection

A router is considered trusted only after:

ROUTER_MATURITY_SEC

If an untrusted router suddenly floods Hellos:

"Unknown router burst"

is raised.

2.13 Stateful Behavioral Detection

Both detectors use:

- sliding windows
- temporal correlation
- neighbor maturity tracking
- protocol validation
- multi-factor scoring

instead of static signature matching.

This improves detection of:

- low-rate attacks
- spoofing

- stealthy reconnaissance
 - protocol abuse
-

3. Overall Detection Pipeline

PCAP Capture



Protocol Parsing



Packet Classification



State Tracking



Sliding Window Analysis



Behavior Correlation



Suspicion Scoring



Alert Generation

4. Security Advantages of the Framework

Lightweight

- No ML dependency
 - Low memory usage
 - Real-time capable
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Protocol Aware

Uses multicast RFC behavior:

- TTL validation
- query timing
- holdtime semantics

- DR election logic
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Stateful

Tracks:

- neighbors
 - multicast groups
 - query history
 - router maturity
-

False Positive Reduction

Uses:

- suspicion scoring
 - membership correlation
 - time-window cleanup
 - trusted router learning
-

THRESHOLDS USED FOR NOW

IGMP Detection Thresholds

Threshold Variable	Value	Purpose
GENERAL_QUERY_IAT_THRESHOLD	100 sec	Minimum allowed interval between IGMP General Queries before considered suspicious
GENERAL_QUERY_BURST_THRESHOLD	3 packets	Number of rapid General Queries needed to trigger flood suspicion
GENERAL_QUERY_WINDOW	300 sec	Sliding time window for General Query flood analysis
GROUP_QUERY_BURST_THRESHOLD	5 packets	Number of Group-Specific Queries required for flood detection
GROUP_QUERY_WINDOW	30 sec	Time window for Group Query flood tracking
MAX_RECENT_MEMBER_EVENT_SEC	10 sec	Correlation window for legitimate membership activity
GROUP_SCAN_WINDOW	30 sec	Time window for multicast group scan detection
MAX_UNIQUE_GROUPS	10 groups	Unique multicast groups accessed before scan alert
QUERIER_MATURITY_SEC	300 sec	Time required before a router becomes trusted
REQUIRE_TTL_1	True	Enforces IGMP TTL validation
TTL Expected	1	RFC-compliant IGMP TTL
Alert Suspicion Threshold	>= 2	Minimum suspicion score required to generate alert

PIM Detection Thresholds

Threshold Variable	Value	Purpose
NORMAL_HELLO_INTERVAL	30 sec	Expected PIM Hello interval
MIN_HELLO_INTERVAL	5 sec	Minimum valid Hello inter-arrival time
WINDOW_SEC	30 sec	Sliding window for Hello flood analysis
MAX_HELLO_COUNT	10 packets	Maximum allowed Hellos inside window
MIN_HOLDTIME	60 sec	Minimum acceptable holdtime
MAX_DR_PRIORITY	1000	Maximum allowed DR priority
MAX_NEIGHBORS	20 neighbors	Neighbor explosion threshold
NEIGHBOR_TIMEOUT	105 sec	Neighbor expiration timeout
ROUTER_MATURITY_SEC	60 sec	Router trust establishment duration
NEW_NEIGHBOR_WINDOW	10 sec	Time window for distributed spoof detection
MAX_NEW_NEIGHBORS	5 neighbors	Maximum new neighbors allowed in short duration
Expected TTL	1	RFC-compliant PIM Hello TTL
Expected PIM Version	2	Valid PIM protocol version
Alert Suspicion Threshold	>= 2	Minimum score required for alert